

Mahmoud Sayed

GitHub

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EDUCATION

- **Misr International University** Cairo, Egypt
Bachelor of Science in Computer Science; GPA: 3.3 Sep. 2024 – Present

EXPERIENCE

- **MSP Tech Club – Software Development Member** Misr International University
Student Club Member Sep. 2024 – Present
 - **Collaboration:** Contribute to club software projects through feature implementation, code reviews, and testing; design and prototype tools used by club members.
 - **Workflow:** Use Git and GitHub for version control following standard branching and PR workflows; maintain project documentation and READMEs.

PROJECTS

- **Van Der Linde – MERN E-Commerce Platform** (*In Development*): Developed a Full-stack luxury watch e-commerce platform built with React 18, Vite, and React Router v6, featuring Context APIs, JWT auth via Axios interceptors, Framer Motion, and Stripe. Backend uses Express/Mongoose with MVC, Google OAuth, and Multer; built in a 5-person team with Git feature branching. **Technologies:** React 18, Node.js, Express.js, MongoDB, JWT, REST API, Stripe, Git.
- **MIU Guide – University Student Portal:** Redesigned the official university website to solve real student pain points: interactive 3D campus map, live GPA calculator, attendance tracker with dropout warnings, and a campus restaurant browser. Features protected routes and a scalable shared component library. **Technologies:** React.js, CSS Modules, Framer Motion, Model-Viewer, React Router.
- **Generalized Tic-Tac-Toe Solver – Algorithms Analysis & Design:** Engineered a generalized Tic-Tac-Toe solver in C++ that computes optimal moves on any NxN board using Minimax, comparing a Brute Force against a Dynamic Programming memoized version with transposition table caching. Benchmarked both implementations across board sizes and search depths, demonstrating measurable performance gains from exploiting overlapping subproblems in adversarial game trees. **Technologies:** C++, Minimax, Brute Force, Dynamic Programming.
- **E-Commerce Management System – OOP:** Desktop application with Admin, Seller, and Customer roles built on a layered OOP architecture with file-based persistence via Java serialization. **Technologies:** Java 11+, JavaFX, OOP, File I/O.
- **Smart Irrigation System – Embedded Systems:** Automated soil-moisture-based irrigation system with two implementations: Arduino (C++) and low-level Assembly (8086), demonstrating hardware control at both abstraction levels. **Technologies:** Arduino, C++, Assembly (8086), Embedded Systems, IoT.

TECHNICAL SKILLS

- **Languages:** JavaScript, C++, Java, Python, SQL, HTML, CSS
- **Technologies:** React.js, Node.js, Express.js, MongoDB, Mongoose, Vite, Axios, React Router, Context API, Framer Motion, JavaFX, Qt, Arduino, MySQL, Git, GitHub
- **Concepts:** REST API Design, OOP, Data Structures & Algorithms, MVC Architecture, JWT Authentication, Minimax & Game Tree Search, Dynamic Programming, Component-Based Design, Problem Solving, Embedded Systems

CERTIFICATIONS

- **Front-End Development Course** MSP Tech Club – Misr International University
HTML, CSS, JS, and Front-End Development Fundamentals 2025
- **Python Essentials 1** Cisco
Python programming fundamentals, problem solving, and core programming concepts Sep. 2025